DIGITAL NUMBER SYSTEMS — EXAM CHEAT SHEET

Core systems: Decimal (10), Binary (2), Octal (8), Hexadecimal (16)

1. Place-Value Rules

Any base b: digits to the left use b^0, b^1, b^2...; digits to the right use b^{-1}, b^{-2}, ...

Decimal value = Σ (digit × b^position)

2. Binary ↔ Decimal

Decimal integer → Binary: repeatedly divide by 2; read remainders bottom → top.

Decimal fraction → Binary: repeatedly multiply fraction by 2; read integer parts top → bottom.

Example: $26.25_{10} = 11010.01_2$

Binary → Decimal: multiply each bit by its power of 2 and add.

$11010.01_2 = 1\times16 + 1\times8 + 0\times4 + 1\times2 + 0\times1 + 0\times2^{-1} + 1\times2^{-2} = 26.25_{10}$

3. Binary ↔ Octal

Shortcut: 1 octal digit = 3 binary bits. Group binary from the radix point outward in groups of 3.

$110\ 101\ 011_2 = 653_8$

$6\ 5\ 3_8 \rightarrow 110\ 101\ 011_2$

4. Binary ↔ Hexadecimal

Shortcut: 1 hexadecimal digit = 4 binary bits. Group binary in groups of 4.

$9A_{16} \rightarrow 1001\ 1010_2$

$1111\ 0110_2 \rightarrow F6_{16}$

Hex	Binary	Hex	Binary	Hex	Binary	Hex	Binary
0	0000	1	0001	2	0010	3	0011
4	0100	5	0101	6	0110	7	0111
8	1000	9	1001	A	1010	B	1011
C	1100	D	1101	E	1110	F	1111

5. Binary Arithmetic

Addition: 0+0=0, 0+1=1, 1+0=1, 1+1=10, 1+1+1=11.

Subtraction: 0−0=0, 1−0=1, 1−1=0; when 0−1 occurs, borrow 1_2 (equivalent to 2 decimal).

Multiplication: ×0 = 0; ×1 = same number; shift left by 1 for ×2.

6. Complement Cheatsheet

Complement	Definition / Method	Example
1's complement	Flip every bit: 0 ↔ 1	101100 → 010011
2's complement	1's complement + 1	101100 → 010011 + 1 = 010100
7's complement	For each octal digit d: 7 − d	$526_8 \rightarrow 251_8$
8's complement	7's complement + 1	$526_8 \rightarrow 252_8$
9's complement	For each decimal digit d: 9 − d	583 → 416
10's complement	9's complement + 1	583 → 417
15's complement	For each hex digit d: F − d	$3A7_{16} \rightarrow C58_{16}$
16's complement	15's complement + 1	$3A7_{16} \rightarrow C59_{16}$

7. Subtraction Using 2's Complement

To calculate A − B: (1) write B in fixed n-bit form; (2) take 2's complement of B; (3) add it to A; (4) discard final carry if present. If no carry, the result is negative and is represented in 2's complement.

Example: 7 − 5 using 4 bits: 0111 + 2's complement(0101). 2's complement(0101) = 1011. 0111 + 1011 = 1 0010 → discard carry → 0010 = 2.

8. Signed Binary & 2's Complement

For n-bit 2's complement: range = −2^(n−1) to +2^(n−1) − 1. For 8 bits: −128 to +127.

Positive numbers: ordinary binary with leading 0. Negative number: positive magnitude → invert bits → add 1.

Sign bit: MSB 0 = non-negative; MSB 1 = negative (for 2's complement).

Overflow: adding two positives gives a negative, or adding two negatives gives a positive → overflow.

9. Decimal ↔ Octal / Hexadecimal

Decimal integer → base b: repeatedly divide by b; read remainders bottom → top.

Decimal fraction → base b: repeatedly multiply by b; read integer parts top → bottom.

Example: $26_{10} \rightarrow 32_{8}$ and $1A_{16}$.

Hex digits: A=10, B=11, C=12, D=13, E=14, F=15.

10. Octal ↔ Hexadecimal

Convert through binary: Octal → Binary (3 bits/digit) → regroup into 4 bits → Hex.

Example: $725_{8} \rightarrow 111\ 010\ 101_{2} \rightarrow 0001\ 1101\ 0101_{2} \rightarrow 1D5_{16}$.

11. Quick Octal Table

Octal	Binary	Decimal
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7

12. High-Value Exam Shortcuts

- 1 hex digit = 4 bits; 1 octal digit = 3 bits; 1 byte = 8 bits.
- Binary ↔ hex: group in 4. Binary ↔ octal: group in 3.
- For grouping fractions, add zeros at the outer end if needed; zeros do not change value.
- 2's complement = invert bits + 1.
- n-bit unsigned range: 0 to 2^n − 1.
- n-bit 2's complement signed range: −2^(n−1) to 2^(n−1) − 1.
- Left shift by k bits ≈ multiply by 2^k when there is no overflow.
- In complements, use the fixed number of digits/bits specified by the question.

13. Mini Practice Set

Question	Answer
9.5_{10} → binary	1001.1_{2}
26.25_{10} → binary	11010.01_{2}
11010.01_{2} → decimal	26.25_{10}
$9A_{16}$ → binary	10011010_{2}
11110110_{2} → hexadecimal	$F6_{16}$
725_{8} → binary	111010101_{2}
101101_{2} → decimal	45_{10}
1's complement of 101001	010110
2's complement of 101001	010111

Golden rule: Integer → divide by the base. Fraction → multiply by the base. Binary ↔ octal/hex → group bits. Complements → subtract from (base − 1), then add 1 for the radix complement.